

PREFACE

This field report has been prepared as part of the academic requirement for the course In Power Plant Equipment under the Department of Electrical Engineering, Pulchowk Campus. The report documents our five day educational visit to Upper Tamakoshi Hydropower Plant 456MW in Dolakha, which was conducted on June 7, 2025.

A group of eight electrical engineering students participated in this field study to gain firsthand knowledge of the design, layout, and operation of a functioning hydropower plant. It was 3 days visit to the power plant and 2 days travelling by road.

The visit provided valuable insights that complemented our classroom learning and enhanced our understanding of real-world power generation systems. We hope this report serves as a useful reference for students and professionals interested in the practical aspects of hydropower generation.

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to the **Department of Electrical Engineering, Pulchowk Campus**, for providing us the opportunity to organize this educational field visit and help us explore the Upper Tamakoshi Hydro power plant.

We would also like to thank Electrical Engineer, **Mr. Jitendra Dev Chhetri** at Upper Tamakoshi Hydropower Plant for his warm hospitality, technical briefings, and guided tour of the facility. His cooperation and willingness to share knowledge made this visit highly informative.

We also appreciate the support and guidance of our faculty members, whose mentorship was instrumental in helping us link theoretical concepts with practical applications during this visit.

Finally, we extend our thanks to all our team members for their active participation and collaborative spirit throughout the trip.

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Introduction

This report presents a summary of the educational visit to the Upper Tamakoshi Hydropower Project (UTKHP) by students from the Department of Electrical Engineering, Pulchowk Campus, Institute of Engineering (IOE). The primary objective of the visit was to gain first hand exposure to the design, construction, and operation of Nepal's largest hydropower project, and to enhance practical understanding of hydroelectric generation systems.

Overview of the Project

Project Type: Run-of-River (with daily peaking storage)

Installed Capacity: 456 MW

Turbine Units: 6×76 MW Pelton-type turbines

Developer/Operator: Upper Tamakoshi Hydropower Ltd. (UTKH)

Commissioned: July 2021 A.D.

River: Tamakoshi River (tributary of Sunkoshi)

Annual Energy Output: ~2,281 GWh (Gigawatt-hours)

Site Description

The project site is situated in the Gaurishankar Rural Municipality of Dolakha District, approximately 194km northeast of Kathmandu. The area features rugged mountainous terrain, steep valleys, and high seismic activity. The access road via Lamosanghu–Charikot–Gongar corridor is narrow but operational, with occasional landslide risks during monsoon season. The Dam located in Lamabager which is approximately 13.4 Km away from the Powerhouse in Gongar. Dam and power house is connected through headrace tunnel of 8.4 km in length

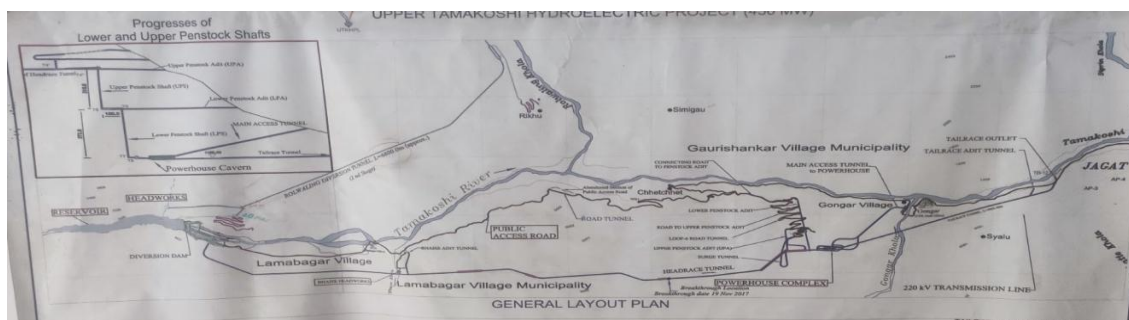


Figure 1: Site Location and General Layout

Components Used In The Power Plants

Reservoir:

The project features a 22-meter-high concrete dam and a live storage capacity of 1.2 million cubic meters. This storage allows for daily regulation of water flow, ensuring consistent power generation during peak hours. The facility is designed with a gross head of 822 meters and a design discharge of 66 cubic meters per second. It comprises six Pelton turbine units housed in an underground powerhouse, contributing to an annual electricity generation of approximately 2,281 GWh.



Figure 2:Reservoir

Dam

The dam is a **concrete gravity structure**, designed to resist the force of the impounded water primarily through its own weight. The dam stands **22 meters tall**, sufficient to create the necessary head for high-pressure water conveyance. It spans **60 meters** across the river, forming a small reservoir that diverts water into the headrace tunnel rather than storing large volumes. Unlike storage dams, this is part of a **peaking run-of-the-river** system. It does not hold significant water but channels it efficiently to maintain a steady flow to the turbines during peak demand. The dam supports a maximum **design discharge of 66 cubic meters per second (m^3/s)**, feeding into the 8.4 km headrace tunnel leading to the Powerhouse.



Figure 3: Diversion DAM and Overflow Spillway

Intake

Intake is the path through which water is diverted into the settling basin from the catchment area. Trash rack at the intake helps filtering the debris present in the water. The water through the intake goes to the settling basin further improving the quality of the water.



Figure 4:: Intake with debris at the entrance blocked by trash rack

Settling basin

Settling basin allows the slit particle to settle down on the floor of the basin before entering into the forebay.



Figure 5: Settling Basin Damaged and Working

There are two numbers of basins each having length approximately 225m in length, 26m in width and depth 17.7m. The settling basin is designed to remove sediments of size 0.15 mm and larger. It operates based on gravity settling, where the velocity of the water is reduced to allow sediment to settle at the bottom.

These basins are critical for protecting the turbines (Pelton-type in this case) from abrasive wear. The settled silt is removed periodically through flushing systems to maintain efficiency.

As can be seen in the figure one of the settling basins got damaged due to a landslide in the area and is under maintenance while the other is in normal operation.

Forebay

The water from the settling basin is entered to the forebay with small storage of water and the water from the forebay gets entered to the tunnel with a trash rack installed at the entry of the headrace tunnel.



Figure 6:Forebay

Headrace Tunnel

The length of the headrace tunnel is 8.4 km with cross-sectional area of 32.14 m².

Surge Tunnel

Surge tunnel connected to the end of the headrace tunnel used to maintain the water pressure in the Penstock shaft and water level during the governing mechanism is active.

Penstock

Penstock takes the water from the headrace tunnel to the turbine through the nozzle. In the power house Shaft penstock was used. The penstock section was divided into upper and lower shafts as shown in the figure. The penstock was divided into two portions because of the construction difficulty during the construction of the shaft. Two penstock shafts are connected through 100m horizontal tunnel.

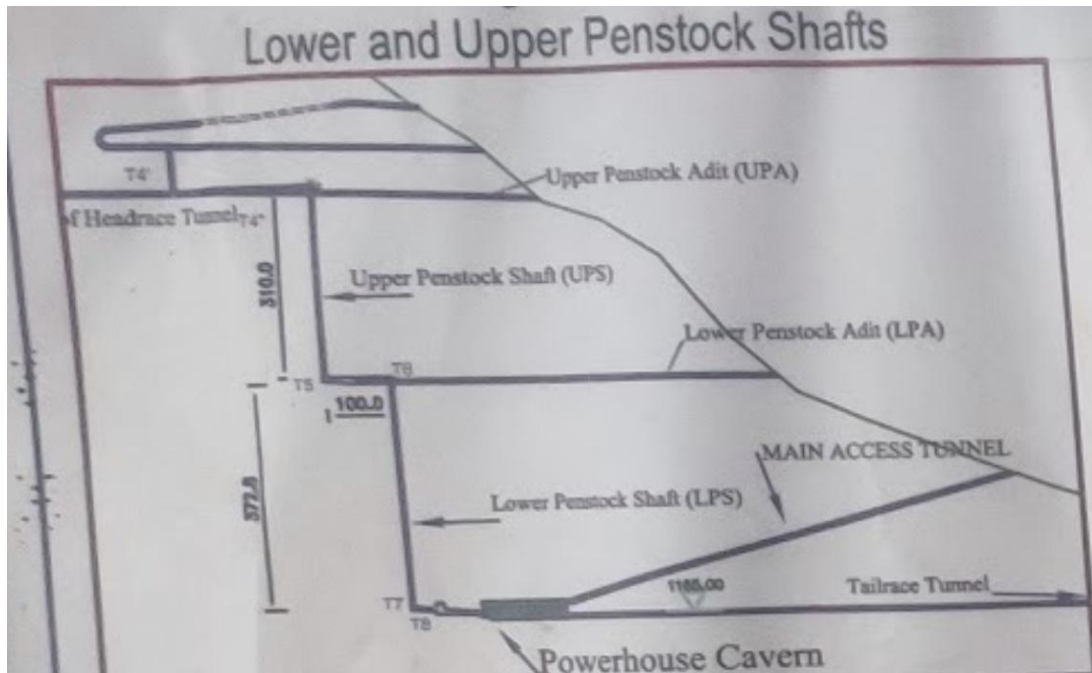


Figure 7: Overview of the penstock Shaft

Main Inlet Valve

All the water from penstock passed through the main inlet valve and then diverted to the 6 secondary inlet valves of each unit. The discharge capacity of the main inlet valve is 66 cubic meters per second. Mostly the main inlet valve is in open position and is closed only when there is need of maintenance work in between the valves.



Figure 8: Main Inlet Valve

Secondary Inlet Valve

This inlet valve provides the direct connection of penstock with the turbine. There are a total of 6 Inlet Valves connected to each of the turbine units. This inlet valve is open during the full operation of the turbine and is closed if maintenance is required in any turbine unit. The following figure shows the closed valve of a turbine unit during maintenance.



Figure 9: Secondary Inlet Valve

Turbine

Turbines are a crucial part of the power house. For a high head power plant, a pelton turbine is used. Pelton turbine is an impulse turbine which converts most of the hydraulic energy into mechanical/kinetic energy. A total of 6 units of peloton turbine is in use. Corrosion of the turbine is a common phenomenon which is caused by the silts present in the water from the headrace tunnel. Regular maintenance is done to ensure the proper quality of the turbine buckets.



Figure 10: pelton turbine

Tailrace Tunnel

Tailrace Tunnel is present downstream of the turbine unit which takes the used water out of the power house into the river. In Upper Tamakoshi Hydro power plant the length of the headrace tunnel was about 2.9 km in length

Coupler

Coupler helps to connect the turbine shaft with the generator shaft. Here in the figure we can see the coupler has been used to connect the shaft of the turbine at the bottom with the upper shaft of the generator.



Figure 11: Coupler in The Power house

Generator

A three phase synchronous generator is used in the power plant. A total of 6 units of generator is used in the plant. Each generator is of 90 MVA capacity with output voltage of 11kv. Andritz Hydro has designed the generator specifically for the upper tamakoshi hydro power plant.

The generator is of silent pole type with rated speed of 600 rpm. More information can be obtained from the figure below.

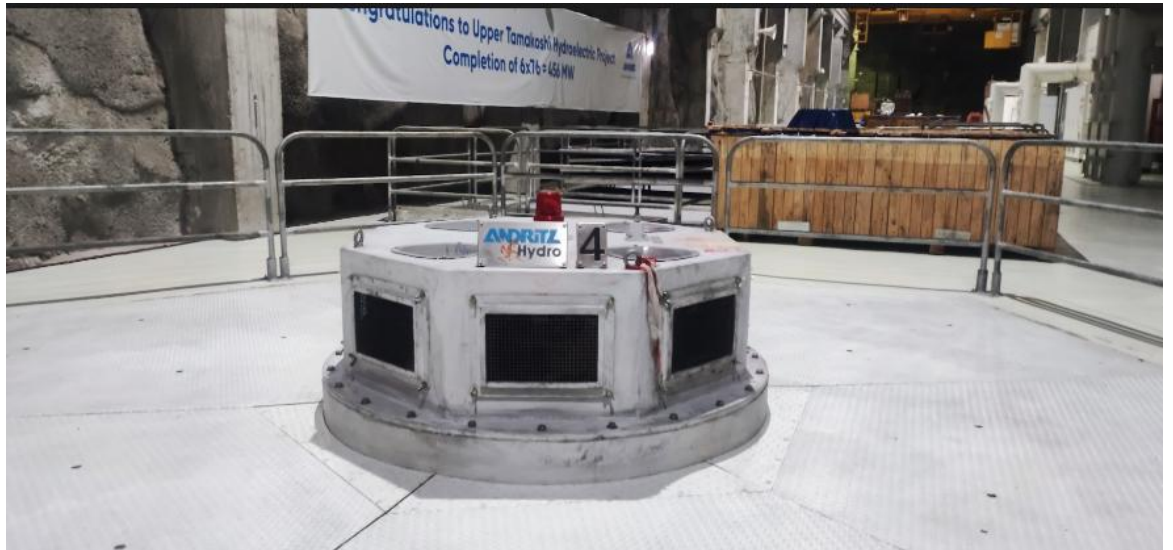


Figure 12: Three Phase Generator

Rated Output	kVA	90000
Rated Voltage	kV	11 ± 5%
Rated Current	A	4723,8
Rated Power Factor		0,85
Rated Frequency	Hz	50
Rated Speed	1/min	600
Permissible Overspeed	1/min	1041
Direction of Rotation (View from Top)		
Rated Field Current	A	1137,1
Rated Field Voltage	V	195,4

Figure 13: Name plate Of Three phase Generator used

Excitation System

Static excitation is used in the generator. Static excitation refers to the excitation of the field using the current generated by the generator itself. Initially the external excitation is provided to the generator through the thyristor bank. Battery backup system is also used for providing the excitation of the field. Diesel generator is also kept as backup generator in case of complete shutdown of the plant. The external source is provided until the excitation reaches up to 30% of its peak excitation after that the field excitation is provided using its

own generated voltage. This phenomenon is known as field flashing. Thyristor bank is used to convert the ac voltage generated into dc for excitation purpose and battery charging purpose. There is separate substation for the diesel generator in the plant which helps to maintain the safety of the generator.



Fig: Figure 14: Diesel generator back up for Excitation

Transformer

Power Transformer:

A single phase power transformer is used in the plant. A total of 19 single phase transformers are present in the plant with 18 in use and 1 spare. 3 of the transformer was used to form the three phases of the transformer. And in this was a total of 6 units of 3 transformers operating in parallel. Each transformer has a power rating of 30 MVA. All of the transformers step up the 11 KV voltage to 230 KV.

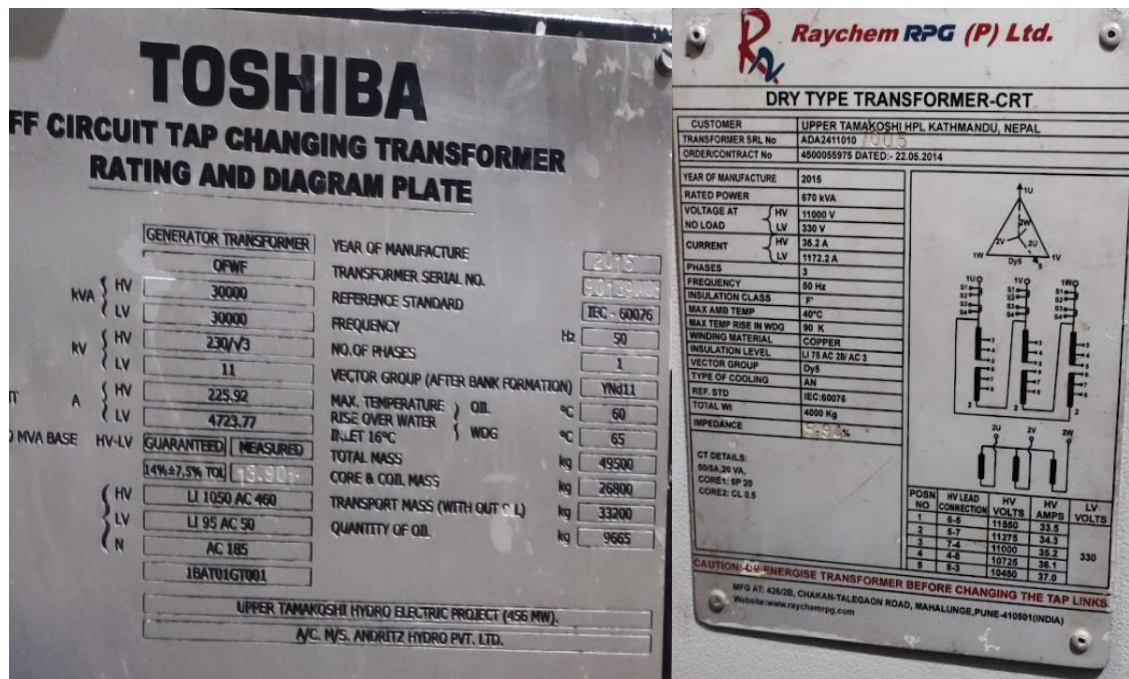


Figure 15: Power Transformer and SST name plate

Substation and Transmission

GIS (Gas Insulated Substation) is used for the switchgear purpose. The use of high transmission voltage i.e about 220 KV make it suitable for the use of GIS. Also, the power house was underground so GIS helped in maintaining the space inside the powerhouse.

220KV transmission line was taken out of the substation which was connected to the transmission switchgear as shown in the figure below. From here it was ready to be transmitted to Khimti Substation from where it was connected to INPS.



Figure 16: Outdoor Switchyard for Transmission

Communication system & SCADA system

There are 3 communication system used in this power plant.

- Walkie Talkie (Radio communication)
- Fiber optics communication (SCADA Communication)
- Nepal Telecom Communication (External Communication) Communication with LDC (Load Dispatch Center)

SCADA means Supervisory Control and Data Acquisition. The SCADA system is an advanced control and monitoring system used in hydropower plants to manage and automate operations. It allows operators to remotely monitor, control, and collect real-time data from various components. Fiber optics has been used for the function of the SCADA system in the plant.

DC source

For the DC Source battery bank is present in the powerhouse which gets its charging through the battery management system. Thyristor bank is also available which converts AC into DC.



Figure 17: DC source

Earthing method

Resistance Transformer Earthing was used in the generator. Bentonite mat was used for increasing the surface area for earthing.

Cooling system And Fire protection

Closed Loop Water cooling system is used in this hydropower plant for cooling of bearing, shaft, generator. Here, river water is used for cooling closed loop water as heat exchanger

For fire hazard protection, one water tank is available there at top floor which provides water supply in case of any fire attacks. The system is also fitted with automatic sprinkler system with sensors for the automation during any hazard safeguarding the plant from human errors.



Figure 18: Closed Loop Water Cooling System



Figure 19: Water Reservoir for Cooling and Fire protection system

Protection system:

For the protection of the generator, transformer and Busbar in the power plant Digital relay are used. Automatic fault detection system and fault elimination is used in the plant using computer algorithms. The control panel for the protection is as shown in the figures below.



Figure 20: Control panel and generator protection scheme



Figure 21: Line Protection and Diff. Cable Protection System

Fire Control Panel:

Fire control panel is shown in the figure below. Monitor the plant situation with the help of the sensors used in the plant. If fire is detected automatic fire extinguishing system get activated. Manual operation is also possible through the control panel.



Figure 22:Fire Control Panel

Breaking mechanism In the Plant:

There can arise a situation where the turbine need to be stoped completely for maintenance purpose or at low load condition. Hence breaking has to be applied to make the turbine motion zero. Initially electrical break is applied by using the residual field. After the Turbine speed has fallen to about 300 rpm mechanical break is applied to completely stop the turbine. Below figure shows the digital monitoring of the breaking system applied in the plant .

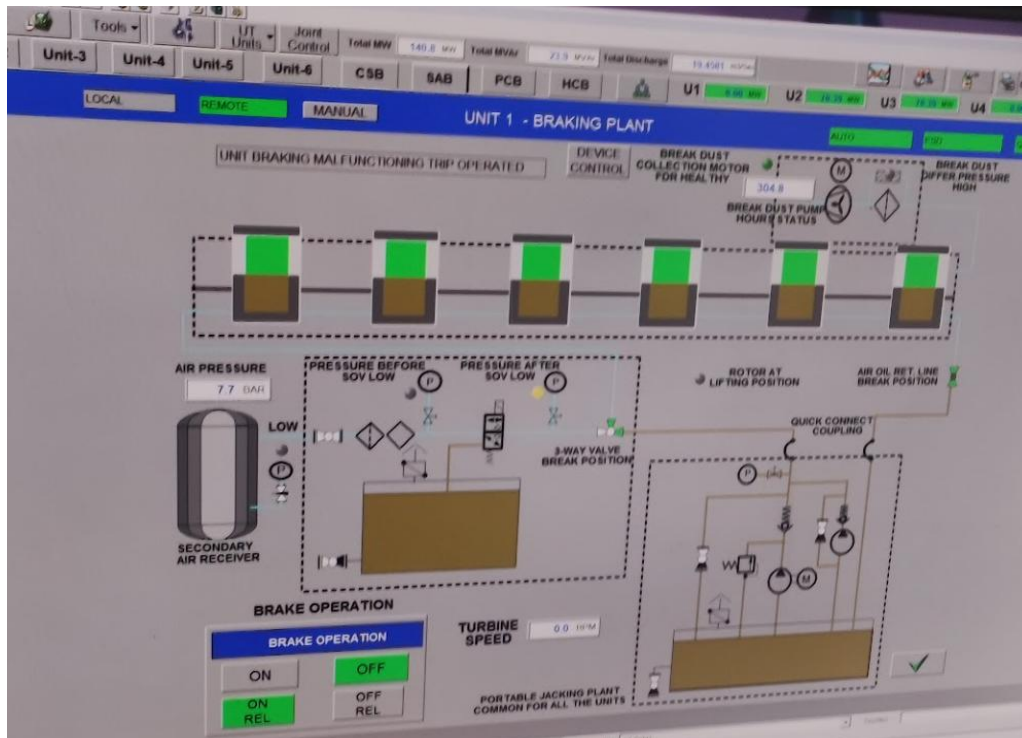


Figure 23:Braking Mechanism

Condition Monitoring Of the Generator system

Condition monitoring of the plant equipment such as generators is very crucial. Because faulty operation of these equipment can lead to the damage of the components and could lead to the halt of its operation. In Upper Tamakoshi Hydro power plant condition monitoring system has been applied of continuous monitoring of the generator performance. The temperature of the different part of the generator is continuously monitored digitally . Similarly the vibration of different part of the generator provides the information about the safe operation of the generator. Hence The vibration monitoring is also applied in the generator.

Below figures shows the data of temperature and vibration of different part of the generator during our visit to the power plant.

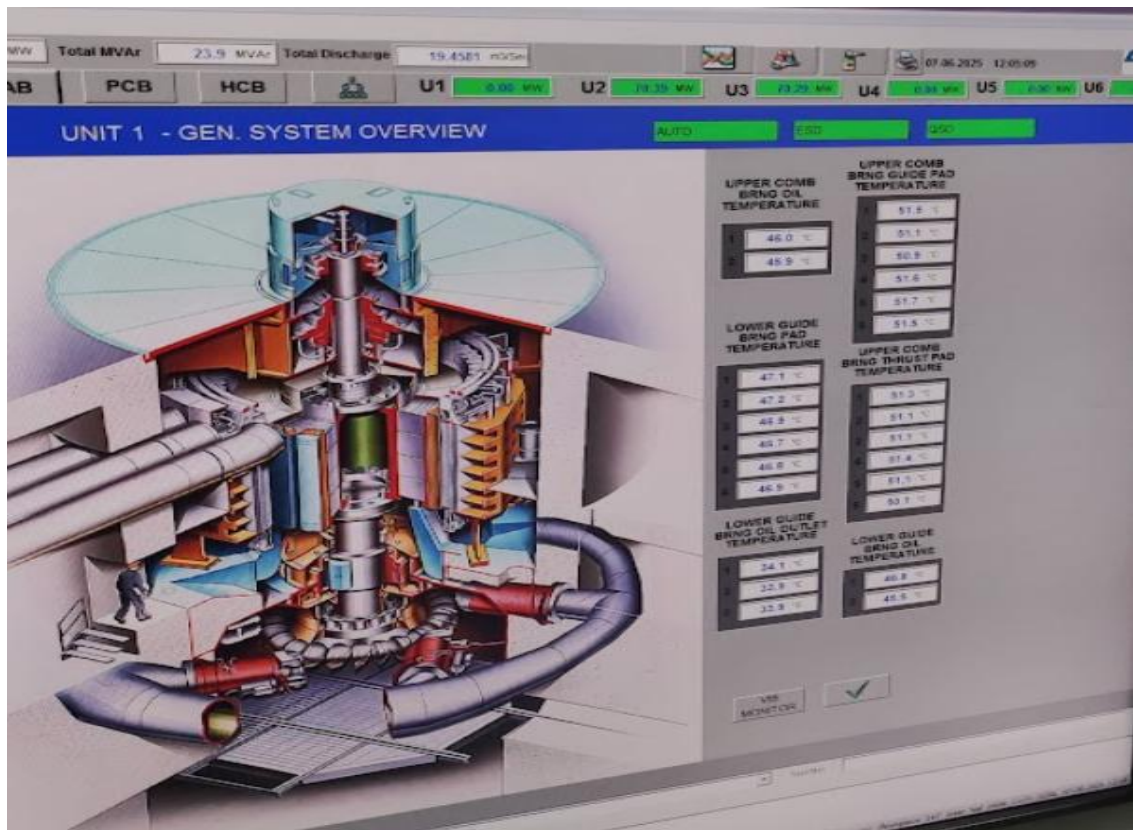


Figure 24: Temperature monitoring

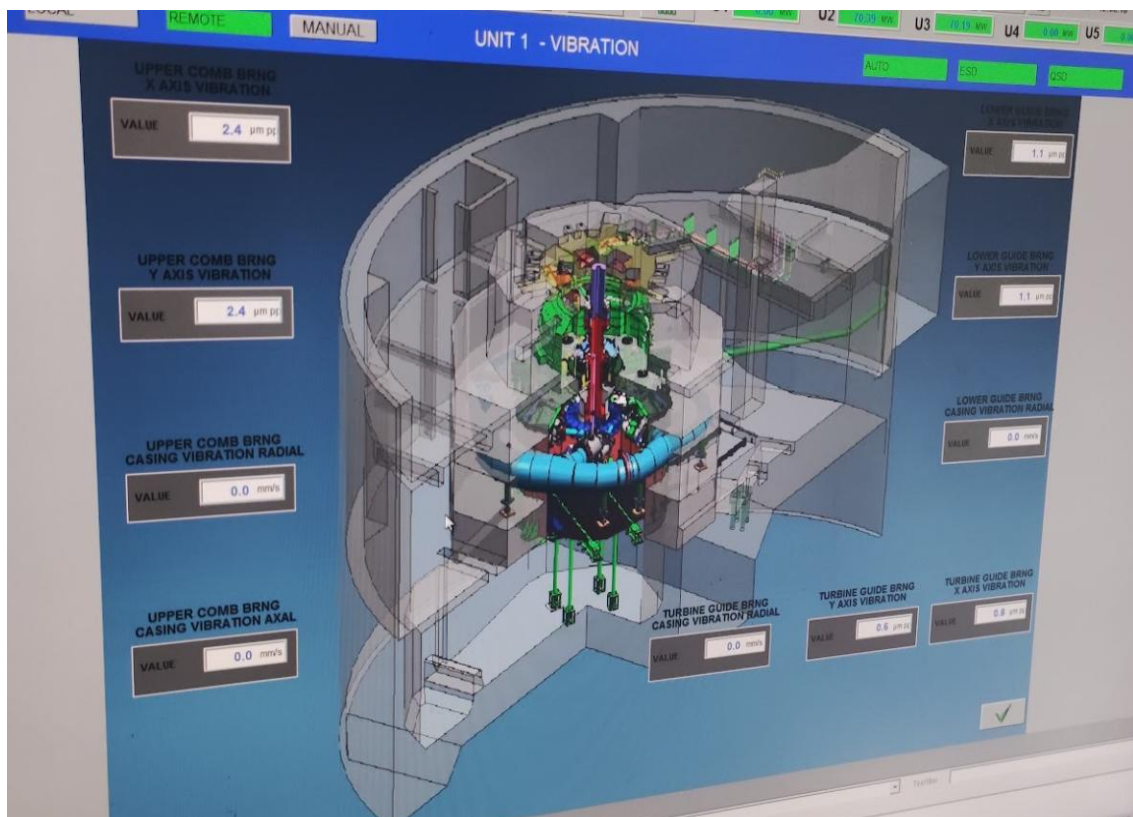


Figure 25: Vibration Monitoring

SCADA System overview:

SCADA refers to Supervisory Control And Data Acquisition. The over all plant Scada system is shown below.



Figure 26: Overall SCADA overview

Daily Load forecasting and actual load data

Load Forecasting is a crucial part of the powerhouse. Based on the previous day's data the plant estimated the demand. The data sheet showing the difference between the forecasted load and actual demand is shown in the fig below. Similarly The water level in the reservoir is also estimated throughout the day and is compared with the actual data throughout the day. The curve is drawn between the energy consumed and the time of the day. Average hourly consumption is calculated and curve is drawn as shown in the figure. Similarly the cure between Reservoir water level and the time of the day is also drawn as shown below.

Hours	Forecasted/Target			Hours	Actual						RWL (m)	Actual Fore RWL
	Ave. Energy (MWh)		RWL (m)		Ave. Ener. (MWh)	D/S Release	Hourly Ave. Inflow	Cum. Ave. Inflow				
	Hrly	Cum.										
Start of Day				Start of Day	Hrly	Ave.	Cum.	m3/s	m3/s	m3/s		
0 00-1 00 (AM)	200	200	1983.57	0 00-1 00 (AM)	283	283	283	1.30	22.8	22.8	1983.60	1983.60
1 00-2 00 (AM)	220	420	1983.58	1 00-2 00 (AM)	260	272	543	1.30	27.2	25.0	1983.20	1982.95
2 00-3 00 (AM)	240	660	1983.54	2 00-3 00 (AM)	203	249	746	1.30	28.6	26.2	1982.90	1982.90
3 00-4 00 (AM)	240	900	1983.49	3 00-4 00 (AM)	163	227	909	1.30	29.0	26.9	1983.00	1983.00
4 00-5 00 (AM)	240	1,140	1983.45	4 00-5 00 (AM)	183	218	1,091	1.30	27.7	27.1	1983.00	1983.00
5 00-6 00 (AM)	240	1,380	1983.40	5 00-6 00 (AM)	208	217	1,299	1.30	27.7	27.2	1982.91	1982.91
6 00-7 00 (AM)	240	1,620	1983.35	6 00-7 00 (AM)	135	205	1,434	1.30	30.3	27.6	1983.12	1983.12
7 00-8 00 (AM)	240	1,860	1983.31	7 00-8 00 (AM)	38	184	1,472	1.30	34.3	28.4	1983.68	1983.68
8 00-9 00 (AM)	240	2,100	1983.26	8 00-9 00 (AM)	42	168	1,514	1.30	33.6	29.0	1984.18	1984.18
9 00-10 00 (AM)	200	2,300	1983.34	9 00-10 00 (AM)	44	156	1,558	1.30	30.9	29.2	1984.60	1984.60
10 00-11 00 (AM)	200	2,500	1983.41	10 00-11 00 (AM)	62	147	1,620	1.40	21.7	28.5	1984.80	1984.80
11 00 AM -12 00 PM	200	2,700	1983.48	11 00 AM -12 00 PM	131	146	1,751	1.40	43.6	29.8	1985.20	1985.20
0 00-1 00 (PM)	150	2,850	1983.70	0 00-1 00 (PM)								1985.30
1 00-2 00 (PM)	120	2,970	1983.99	1 00-2 00 (PM)								1985.60
2 00-3 00 (PM)	120	3,090	1984.27	2 00-3 00 (PM)								1985.85
3 00-4 00 (PM)	130	3,220	1984.52	3 00-4 00 (PM)								1986.06
4 00-5 00 (PM)	250	3,470	1984.45	4 00-5 00 (PM)								1986.01
5 00-6 00 (PM)	300	3,770	1984.25	5 00-6 00 (PM)								1985.84
6 00-7 00 (PM)	270	4,040	1984.13	6 00-7 00 (PM)								1985.74
7 00-8 00 (PM)	320	4,360	1983.87	7 00-8 00 (PM)								1985.53
8 00-9 00 (PM)	330	4,690	1983.57	8 00-9 00 (PM)								1985.27
9 00-10 00 (PM)	300	4,990	1983.35	9 00-10 00 (PM)								1985.09
10 00-11 00 (PM)	240	5,230	1983.30	10 00-11 00 (PM)								1985.05
11 00 PM -12 00 AM	240	5,470	1983.26	11 00 PM -12 00 AM								

Figure 27:Load Data Sheet

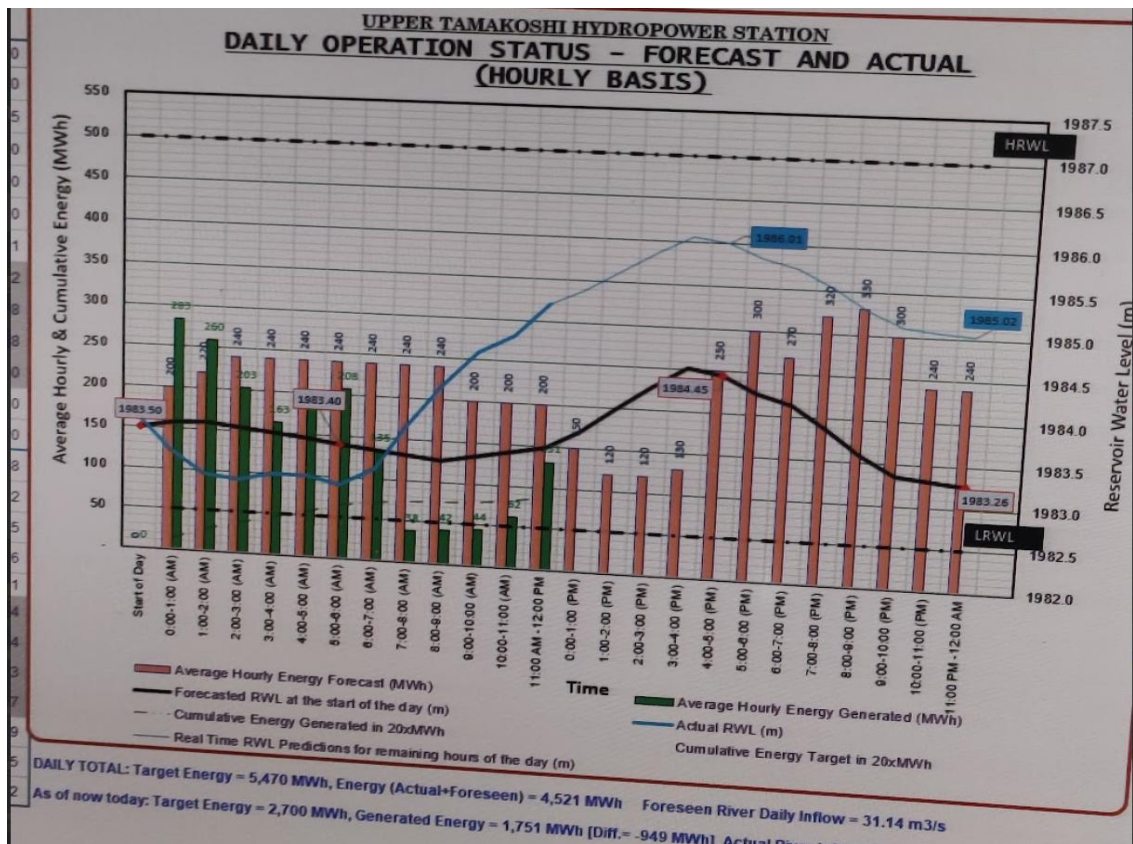


Figure 28:Daily load Forecasting curve

Overall SLD of the plant:

The overall single line diagram of the plant shows all the equipment used in the plant. It includes Transformers, Generators, Substations and busbars. Isolators and Circuit breakers also involved in the single line diagram. By Observing a single line diagram of a plant an electrical engineer should be able to understand the overall plant operation. He/She should be able to tell how many generating units is being used in the plant, how many transformers are there, the voltage level of the system, type of busbar used. From the above SLD we can see that there are 6 generation units used in the power plant. Among six units two units are in operation. Tapping is done form the 11kv line for SSt is also visible. There are circuit breakers and isolators used at different parts of the plant. We can see that circular icon are representing isolators while square shape icon represents circuit breaker. Double Bus single break bus bars are being used. Also double circuit XLPE aluminum cable is being used for transmission purposes.

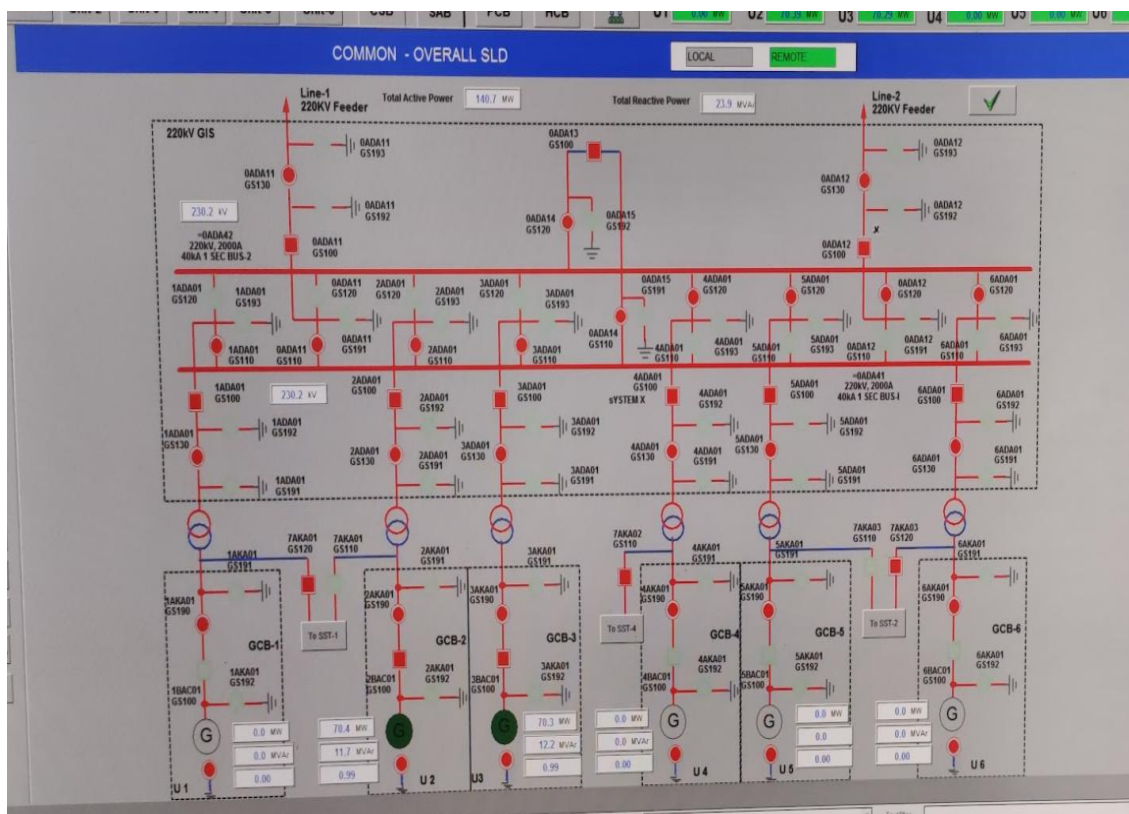


Figure 29:SLD of Upper Tamakoshi Hydro power plant

Discussion:

Visiting the Largest power plant (**i.e Upper Tamakoshi Hydropower plant 456MW**) in the nation was a great experience both from a learning perspective and an experience perspective. Our visit provides us a new perspective to look at electrical engineering and powerhouse construction. We gain practical experience about the size of the machines used in the plant. The powerplant equipment all worked together to achieve a common goal of transmitting energy hundreds of kilometers away from the generating station to people's homes. Hundreds of Components balanced by the engineering principles at the exact location was able to generate 456 megawatt of electricity which light peoples homes.

The civil construction involved the dam site which was a gravity Dam storing approximately 1.2 million cubic meters of water, enough for the operation of the power plant continuously. Two settling basins along with a forebay provides quality water to the 8.4 km headrace tunnel. The headrace tunnel is connected to the penstock at its end. The water is rushed into the penstock shaft controlled by the primary main inlet valve and secondary main inlet valve. There are six units of penstock pipe connected to six turbines. 5 nozzles control the output power of the turbine. All controlled digitally based on the demand requirement. A digital governing mechanism is used for controlling the power output. Each generator generating 90 MVA of Power connected with 3 units of single phase transformer each of 30 MVA provides a stepping voltage from 11KV to 230 KV, ready to be transmitted. GIS provides safety to the bus system, transformer and generating units. Closed loop cooling systems maintain the temperature of the generator units. Finally the 230 KV line is taken out at the starting of the entrance tunnel from which it is transmitted from switchyard through transmission line 47 KM away to Khimti substation ready to be connected to INPS.

Conclusion:

Visit To the Upper Tamakoshi Hydropower Plant offered us a clear understanding of hydroelectric generation, system operation, and grid contribution. It made our engineering journey more practical and joyful. It helped us to bridge the gap between the theoretical knowledge gained through our institution to the practical knowledge available in the field. We are now more confident in handling the practical engineering system. Hence this visit completely fulfilled our objective